

- 5 (a) (i) concentric circles, anticlockwise(*minimum 3 circles*)M1
 separation of lines increases with distance from wire A1 [2]
- (ii) direction from Y towards X A1 [1]
- (b) (i) flux density at wire Y = $(4\pi \times 10^{-7} \times 5.0) / (2\pi \times 2.5 \times 10^{-2})$ C1
 $= 4.0 \times 10^{-5} \text{ T}$ C1
 force per unit length = BI
 $= 4.0 \times 10^{-5} \times 7.0$ C1
 $= 2.8 \times 10^{-4} \text{ N}$ A1 [4]
- (ii) *either* force depends on product of the currents in the two wiresM1
 so equal A1
or (isolated system so) Newton's 3rd law applies (M1)
 so equal (A1) [2]

[Total: 9]

Page 4	Mark Scheme: Teachers' version	Syllabus
	GCE A/AS LEVEL – October/November 2009	9702

6 (a) (i) $\omega m f$ induced proportional / equal to